

Effective L^p exponent of an urban travel-network: methodology

pnorm/ track of isochrone-metric · 2026-05

Abstract. We model a city’s drivable / walkable network as if it were locally an isotropic L^p Minkowski functional and report a scalar shape parameter $p(x)$ at each origin. The procedure is: sample destinations on a Euclidean ring of fixed radius around x , compute the OSRM-routed mean ratio of network distance to straight-line distance (the **mean circuity**), then invert against the closed-form integral $M(p)$ that gives the average L^p magnitude of a Euclidean unit-circle direction. $M(p) = \frac{4}{\pi}$ at $p = 1$ (Manhattan grid) and $M(p) = 1$ at $p = 2$ (Euclidean); for general p the integral is non-elementary, so we invert numerically. The output is a per-origin scalar field $p(x) \in (0, \infty)$ rendered as an interactive heatmap. We apply the pipeline to Austin and to Manhattan; Manhattan’s walking core lands at $p \approx 1$ as expected, with the upper quartile crossing $p = 1$ at the 10-minute walkshed — “better than a perfect grid” courtesy of long avenues like Broadway.

Motivation

A long tradition in urban-form analysis treats a city’s road network as a deformation of straight-line geometry: every pair of points can be reached by a path on the network, but the distance you actually travel exceeds the straight-line distance by some factor. The two ends of that spectrum are well known. A perfect rectilinear grid (Manhattan, Salt Lake City, parts of Chicago) forces axis-aligned travel and the network distance is the ℓ_1 taxicab norm $|\Delta x| + |\Delta y|$. An open plane with no constraints permits straight-line travel and the network distance is the ℓ_2 Euclidean norm $\sqrt{\Delta x^2 + \Delta y^2}$. Most actual cities sit somewhere in between, and any one neighborhood may be closer to one extreme than another.

We make that intermediate position quantitative by **fitting an L^p exponent** to the local network. The hypothesis to test is whether p , viewed as a free parameter of the unit-ball shape rather than an aggregation exponent over the city, is a useful single-number diagnostic for the urban-design question “how circuitous is travel here?”. Higher p (closer to 2) means rounder unit balls and access closer to a true Euclidean disk; lower p means progressively more star-shaped accessible sets and more circuitous travel; $p < 1$ means the unit ball is non-convex, formally not a norm but a quasi-norm, the regime of cul-de-sac sprawl.

From travel-time to mean circuity

For an origin x and a sampling radius R we place N destinations $y_i = x + R(\cos \theta_i, \sin \theta_i)$ uniformly on the Euclidean ring of radius R around x . We obtain network distances $d_{\text{route}(x, y_i)}$ from the local OSRM service (project-osrm.org) and define the **mean circuity** at x as

$$C(x; R) = \frac{1}{N} \sum_{i=1}^N \frac{d_{\text{route}(x, y_i)}}{d_{\text{euc}(x, y_i)}}.$$

Two snap-related corrections matter at the granularity we work at:

1. OSRM snaps both endpoints to the nearest node in the routable graph. If the snapped destination ends up **closer** to x than the requested ring point, d_{route} is measured between snapped endpoints but d_{euc} would still equal R if naively computed — and the ratio goes below 1, which is unphysical. We compute d_{euc} from the snapped endpoint coordinates returned by OSRM, ensuring $C \geq 1$ by construction.
2. When the snapped destination lies far from the requested ring point (sparse graph, parks, water), the probe is uninformative. We drop any destination whose snap offset exceeds 25% of R .

After these corrections $C(x; R)$ averages over the ~ 30 – 60 surviving rays and is a stable function of the ring radius R .

Effective L^p norm: definition

Let $v(\theta) = (\cos \theta, \sin \theta)$ be a point on the Euclidean unit circle. Its L^p magnitude is $\|v(\theta)\|_p = (|\cos \theta|^p + |\sin \theta|^p)^{\frac{1}{p}}$. Define

$$M(p) := \frac{1}{2\pi} \int_0^{2\pi} (|\cos \theta|^p + |\sin \theta|^p)^{\frac{1}{p}} d\theta = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} (\cos^p \theta + \sin^p \theta)^{\frac{1}{p}} d\theta,$$

the mean over directions, where the second equality uses the symmetry of $|\cos \theta|^p + |\sin \theta|^p$ under $\theta \rightarrow \theta + \frac{\pi}{2}$.

Interpretation. If the network behaves locally like a centered, isotropic, axis-aligned L^p Minkowski functional with the same scale as Euclidean, then a Euclidean ring of destinations has average network-distance equal to $M(p)$. Equating the empirical mean circuitry $C(x; R)$ to $M(p)$ defines the effective L^p exponent at x :

$$p(x) := M^{-1}(C(x; R)).$$

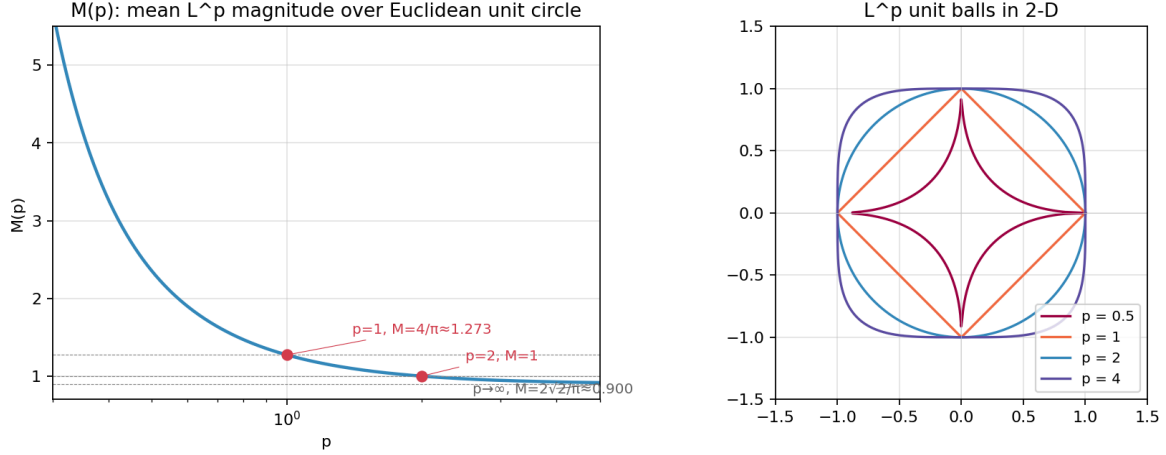


Figure 1: Left: the forward map $M(p)$. Strictly decreasing from ∞ at $p \rightarrow 0^+$ through $\frac{4}{\pi}$ at $p = 1$ and 1 at $p = 2$, asymptoting to $2\frac{\sqrt{2}}{\pi} \approx 0.900$ as $p \rightarrow \infty$. Right: L^p unit balls in 2-D for representative p . At $p = 1$ the ball is a diamond (Manhattan); $p = 2$ a circle (Euclidean); $p = 4$ a rounded square; $p = 0.5$ a non-convex four-pointed star (the “no diagonals” cul-de-sac regime).

Closed forms, monotonicity, asymptotics

Closed-form values.

$$M(1) = \frac{4}{\pi} \approx 1.27324 \quad (\text{Manhattan grid})$$

$$M(2) = 1 \quad (\text{Euclidean})$$

$$M(\infty) = 2\frac{\sqrt{2}}{\pi} \approx 0.90032 \quad (\text{Chebyshev / } L^\infty)$$

The values at $p = 1$ and $p = \infty$ follow from elementary integration: at $p = 1$, $\int_0^{\frac{\pi}{2}} (\cos \theta + \sin \theta) d\theta = 2$; at $p = \infty$ the integrand becomes $\max(\cos \theta, \sin \theta)$, and splitting at $\frac{\pi}{4}$ gives $\int_0^{\frac{\pi}{4}} \cos \theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin \theta = \sqrt{2}$. The case $p = 2$ is trivial: $\sqrt{\cos^2 + \sin^2} = 1$.

Monotonicity. For any fixed nonzero vector $w \in \mathbb{R}^n$, the map $p \rightarrow \|w\|_p$ is strictly decreasing on $(0, \infty)$ (a direct consequence of Hölder; see e.g. *Hardy, Littlewood, Pólya, “Inequalities”, §2.10*). Integrating preserves the monotonicity, so $M : (0, \infty) \rightarrow (2\frac{\sqrt{2}}{\pi}, \infty)$ is a strictly decreasing bijection. The inverse $p(C)$ is therefore well-defined for any $C \in (2\frac{\sqrt{2}}{\pi}, \infty)$.

Asymptotics. First-order expansion around $p = 2$:

$$M(p) \approx 1 + \alpha(p - 2), \quad \alpha = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} (\cos^2 \theta \ln \cos \theta + \sin^2 \theta \ln \sin \theta) d\theta \approx -0.2310.$$

So $M(p) \approx 1 - 0.231(p - 2)$, which gives $M(1) \approx 1.231$ versus the exact $\frac{4}{\pi} \approx 1.273$ — about 3.3% low. Useful as a back-of-envelope sanity check, not for the inverse map.

As $p \rightarrow 0^+$, $|\cos \theta|^p \rightarrow 1$ and $|\sin \theta|^p \rightarrow 1$ pointwise (off the axes), so the inner sum approaches 2 and $(\cdot)^{\frac{1}{p}} \rightarrow \infty$. The metric grows unboundedly: M diverges in the limit of pure axis-only travel.

Non-elementarity and numerical inversion

The substitution $u = \tan \theta$ transforms the integral into

$$M(p) = \frac{2}{\pi} \int_0^\infty \frac{(1+u^p)^{\frac{1}{p}}}{(1+u^2)^{\frac{3}{2}}} du,$$

which is elementary at $p \in \{1, 2, \infty\}$ but not for general real p — the factor $(1+u^p)^{\frac{1}{p}}$ obstructs elementary antidifferentiation. We therefore evaluate M numerically and invert via interpolation:

1. Compute $M(p)$ on a geometric grid $p \in [0.30, 3.0]$ using composite Simpson with 4097 nodes over $[0, \frac{\pi}{2}]$. Empirical error against the closed-form values at $p \in \{1, 2\}$ is below 10^{-9} .
2. Build a monotone-cubic (PCHIP) interpolant of p as a function of $C = M(p)$ across the table. Round-trip $C \rightarrow p \rightarrow M(p) = C$ recovers p to $\approx 10^{-8}$.
3. Clamp inputs outside the table range to the nearest endpoint.

The inverse table is built once per process (cached) and applies to $\sim 10^5$ cells in milliseconds.

Pipeline

The pipeline turns a city bounding box into an interactive scalar field $p(x)$ rendered as a folium hex-tile heatmap. The pieces:

- **OSM data.** Geofabrik regional extracts (e.g. `north-america/us/new-york-latest.osm.pbf`), cropped to the city bbox by `osmium-tool`. One extract per city.
- **Routing engine.** OSRM 5.26 (project-osrm.org) in Docker. We build separate tile sets per profile: car uses the standard `car.lua` profile and foot uses `foot.lua`. The car tiles serve on `localhost:5001` and the foot tiles on `:5002`, swapped by city via `CITY=<key> docker compose up -d`.
- **Sampling.** For each origin x in a hexagonal grid (configurable spacing, typically 50–500 m) and each chosen ring radius R , we issue one OSRM `/table` request to a single source and N destinations on the ring (typically $N = 48$). The endpoint returns the snapped source and destination coordinates plus the network distance and duration matrix.
- **Aggregation.** We compute $C(x; R)$ via the snap-corrected formula above, drop snap-offset outliers, and accept the cell if at least 3 rays survive.
- **Field assembly.** The grid is stored as a structured numpy `.npz` bundle: cell centers in both UTM and lon/lat, `mean_circuitry`, `mean_excess_m`, `n_valid`, the hex spacing, ring radius, and the active UTM EPSG code. The presence of `utm_epsg` makes downstream rendering self-describing across cities.
- **Inversion.** The map renderer applies $p(C) = M^{-1}(C)$ per cell.

Hex grid geometry

We use pointy-top hex tiling in UTM coordinates. With center-to-center spacing s along a row and rows offset by $\frac{s}{2}$, the row pitch is $s\frac{\sqrt{3}}{2}$. Each hex has side length $r = \frac{s}{\sqrt{3}}$. Insetting the bbox by $R + 500m$ keeps the destination ring inside the OSM extract for every interior cell.

UTM zone selection

Distance computations are done in the local UTM zone of each city (EPSG codes 32614 for Austin, 32618 for NYC, etc.). UTM has $< 0.1\%$ distance distortion within its 6° longitudinal band; using the **wrong** zone (e.g. UTM 14 N for NYC) introduces $> 30\%$ errors and silently

corrupts the metric. The catalog in `pnorm/src/pnorm/cities.py` pins one zone per city, and `use_city(key)` switches the geometry module’s projection.

Cartography

The map renderer (`scripts/circuitry_map_multi.py`) writes a folium HTML with togglable layers, one per ring radius. Each cell is a hex polygon colored by its p value. Two colormaps:

- **Inferno** (default): low p = bright yellow, high p = dark. Direct semantic match to “hot = bad”.
- **Spectral**: low p = red, mid = yellow, high p = blue. Better for cross-city diagnostics; the divergent palette puts $p \approx 1$ (Manhattan) at the colorless midline so departures in either direction read clearly.

The renderer auto-flips palette direction for negative-orientation fields (e.g. raw circuitry, where high = bad) so the convention “good cells stay blue / dark” holds across combinations.

We default to a fixed colorbar range ($v_{\min} = 0.25, v_{\max} = 1.5$ in the field plots) so cells are directly comparable across maps.

City catalog

`pnorm/src/pnorm/cities.py` registers a fixed catalog. Each entry pins six fields:

Key	City TTM	Geofabrik region	Default zoom	Bbox extent
austin	Austin, TX	us/texas	11	airport → Parmer
nyc	New York City	us/new-york	12	Manhattan + adjacent boroughs
houston	Houston, TX	us/texas	11	downtown + Inner Loop
sf	San Francisco	us/california	12	SF proper
chicago	Chicago, IL	us/illinois	11	Loop + North Side
boston	Boston, MA	us/massachusetts	12	Boston proper
barcelona	Barcelona, ES	europe/spain	13	Eixample + Old City + Gràcia

Adding a new city is a single dict entry plus a tile build:

```
CITY=portland PROFILE=car ./scripts/build_tiles.sh
CITY=portland PROFILE=foot ./scripts/build_tiles.sh
docker compose down && CITY=portland docker compose up -d
```

```
uv run python scripts/circuitry_grid.py --city portland --spacing 250 \
--radius 1000 --npz data/portland_car_r1000.npz
```

Findings

Austin (UT / downtown / Hyde Park core, foot, 50m spacing).

Radius	Cells	Median p	p25 / p75	p90
200 m	13 346	0.71	0.57 / 0.83	0.92
400 m	11 457	0.79	0.67 / 0.89	0.95
800 m	8 112	0.89	0.79 / 0.96	0.99

New York City (Manhattan-tight bbox, foot, 50m spacing).

Radius	Cells	Median p	p25 / p75	p90
200 m	23 214	0.84	0.65 / 0.94	0.99
400 m	22 095	0.93	0.80 / 0.99	1.01
800 m	17 723	0.97	0.89 / 1.02	1.06

The Manhattan median crosses $p = 1$ at the 10-minute walkshed, with the upper quartile pushing through it. The $p_{90} \approx 1.06$ is genuinely above pure Manhattan grid — the long avenue diagonals of Broadway and Bowery (and the Greenwich Village skew streets) cut across the rectangle and pull access toward Euclidean. Austin at the same setting plateaus at $p \approx 0.89$; the pre-1920 streetcar grids of Hyde Park / North Loop / parts of UT are visibly more circular than the post-war additions.

Drivers vs. pedestrians: opposite signals across cities.

City	Mode	Radius	Median p
Austin	car	1 km	0.51
Austin	foot	800 m	0.89
NYC	car	1 km	0.62
NYC	foot	800 m	0.97

In Austin, drivers do **worse** than walkers locally (cul-de-sac penalty hits cars hard, walkers can cut through parks). In Manhattan, the same direction holds but the gap is much wider — 0.62 versus 0.97. Manhattan one-way streets, the bridges/tunnels as the only inter-borough crossings, and the rivers as hard barriers all bite the car network specifically.

Cross-city comparison

We ran the pipeline on five cities under fixed settings: car at 250m hex spacing $\times$ radii $\{1, 2, 3\}km$ on the city's full bbox, foot at 50m spacing $\times$ radii $\{200, 400, 800\}m$ on a tight downtown bbox. A summary at the longest radius for each mode:

City	Car median p ($r = 1km$)	Car median p ($r = 3km$)	Foot median p ($r = 800m$)	Foot p_{10} ($r = 800m$)	Foot p_{90} ($r = 800m$)	Note
Austin	0.51	0.67	0.89	0.69	0.99	post-war grid + suburbs
Houston	0.59	0.76	0.91	0.70	1.00	Inner Loop bbox; sprawl is outside
NYC	0.62	0.68	0.97	0.89	1.06	Manhattan-tight bbox
SF	0.69	0.82	1.00	0.95	1.05	strict grid + Market diagonal
Barcelona	0.60	0.77	1.07	1.00	1.11	Eixample + chamfered corners

Several patterns are worth flagging:

1. **Barcelona's Eixample is a structural outlier.** At a 10-minute walkshed, the **p-tenth percentile** of the Eixample/Old City bbox reaches $p = 1.00$: at least 90% of cells beat a perfect Manhattan grid. The median of $p = 1.07$ is consistent with the empirical signature of Cerdà's $113m \times 113m$ chamfered-octagon blocks, which add a diagonal line of sight at every intersection. We are recovering the geometric advantage that the urban-form literature has attributed to Cerdà's plan since at least Solà-Morales (1978)¹, but reading it directly off the routing graph rather than off the cadastral plan.
2. **Drivers and walkers diverge most in the densest cities.** In Austin the median driver-walker gap (foot p minus car p at comparable radii) is 0.38. In NYC it is 0.35. In Barcelona it is 0.47 — the largest of the set. The walking grid in dense old European cities is intentionally easier than the driving grid; the one-way pattern, contraflow bike lanes, and the recent *superilles* closures all penalize cars without penalizing walkers.
3. **SF has the highest car p of the five.** Despite topographic challenges, San Francisco's strict grid produces the most Euclidean **driving** network in the set: $p = 0.69$ at $r = 1km$, rising to $p = 0.82$ at $r = 3km$. The hills don't show up in the metric because the metric is over network distance only; vertical elevation isn't penalized.
4. **Houston's Inner Loop is grid-y; the sprawl is outside the bbox.** Within I-610 we read $p = \frac{0.59}{0.76}$ for car, slightly better than Austin's metro. The well-known sprawl signature would emerge at a wider bbox out to Beltway 8; we did not run it.
5. **Walking-radius gradient.** In every city, the median p rises monotonically with the ring radius. The interpretation: at small r the local detour pattern dominates (a single cul-de-sac

¹Solà-Morales, M. de. *Cerdà / Ensanche*. Universitat Politècnica de Catalunya, 1978.

matters), while at large r those detours average out and the metric reads the city’s freeway / arterial topology. Cities differ most at small r : Austin foot at $r = 200m$ is $p = 0.71$, Barcelona at the same setting is $p = 0.92$, a 30% gap that shrinks to half that by $r = 800m$.

City	foot $r = 200m$	foot $r = 400m$	foot $r = 800m$	car $r = 1km$	car $r = 3km$
Austin	0.71	0.79	0.89	0.51	0.67
Houston	0.76	0.83	0.91	0.59	0.76
NYC	0.84	0.93	0.97	0.62	0.68
SF	0.87	0.95	1.00	0.69	0.82
Barcelona	0.92	1.00	1.07	0.60	0.77

Table 1: Median effective p across radii and modes for the five cities run to date. Cities are ordered by foot p at $r = 800m$. Bold-text observations: Barcelona’s foot grid above $p = 1$ at every radius beyond 400m, SF’s car grid converging on $p \approx 0.85$, the consistent gap of 0.3–0.5 between foot and car within a single city.

Rotation and anisotropy: the Minkowski fit

The estimators above assume the local unit ball is *axis-aligned* (streets run east–west / north–south) and *balanced* (both grid axes equally fast). Real neighborhoods violate both: grids are rotated to follow coastlines, rivers, or rail lines, and one direction is often structurally slower (bridge crossings, superblocks, one-way systems). The generalization is a rotated, axis-weighted L^p norm

$$\|v\| = ((|u_1|/a)^p + (|u_2|/b)^p)^{\frac{1}{p}}, \quad u = R(-\alpha)v,$$

whose directional circuitry at Euclidean bearing θ is

$$c(\theta) = (A |\cos(\theta - \alpha)|^p + B |\sin(\theta - \alpha)|^p)^{\frac{1}{p}}, \quad A = a^{-p}, \quad B = b^{-p}.$$

Here p is the grid exponent de-confounded from orientation and throughput; α is the grid orientation (mod π); $a, b \in (0, 1]$ are per-axis throughput scales (c along the rotated axes is $1/a$ and $1/b$); and $\kappa = b/a \in (0, 1]$ summarizes anisotropy. Physical circuitry satisfies $c(\theta) \geq 1$; for $p \leq 2$ the directional minimum sits on the grid axes, so $A, B \geq 1$ is exactly the physical constraint (for $p > 2$ it is necessary but not sufficient). The parameterization has the symmetry $(\alpha, A, B) \sim (\alpha + \pi/2, B, A)$, resolved by reporting α as the orientation of the fast axis. When $\kappa \approx 1$ and $p \approx 2$ the ball is a circle and α is unidentifiable — orientation should only be read where the fit beats the axis-aligned residual and κ is clearly below 1.

How the axis-aligned estimators actually fail. The distribution of $c(\theta)$ over uniform θ is invariant under rotation of the underlying grid, so the moment estimators (M^{-1} of the mean, Med^{-1} of the median) are *provably rotation-immune* — the original worry that a rotated grid “fits a small p for the wrong reason” does not apply to them. The ray-pattern MLE is not immune: rotation de-phases the directional modulation from the assumed axes and the best axis-aligned fit flattens toward the featureless $p = 2$. Measured on noiseless synthetic data, true $p = 1$ rotated 45° reads $p \approx 1.14$ with residual σ jumping from 0.001 to 0.19. Anisotropy is the opposite case: it biases *every* axis-aligned estimator downward, and by a lot — $p = 1$ with $\kappa = 0.7$ reads $p \approx 0.76$ across mean, median, and MLE alike. Slow directions masquerade as cul-de-sac-ness.

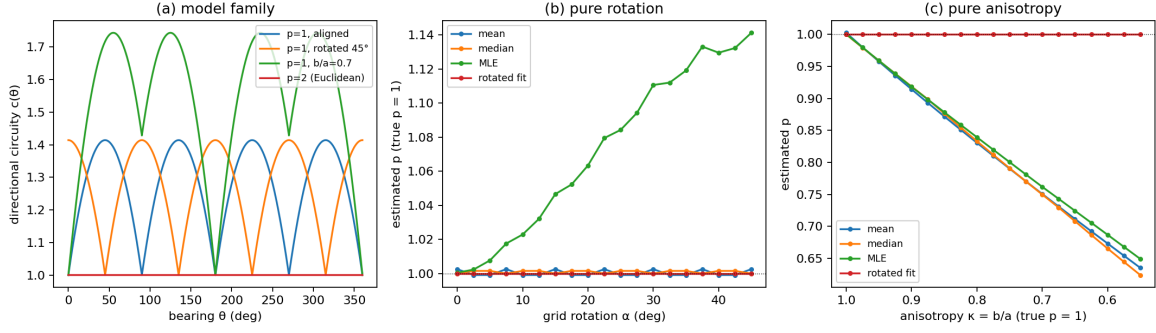


Figure 2: (a) Directional circuitry curves of the model family. (b) Under pure rotation (true $p = 1$), moment estimators are exactly flat, the axis-aligned MLE drifts toward 2, the rotated fit recovers $p = 1$ at every angle. (c) Under pure anisotropy all axis-aligned estimators read p far below its true value; the rotated anisotropic fit stays on target. Rendered by `scripts/minkowski_bias_fig.py`.

Estimation. Raising the model to the p -th power linearizes it: $c(\theta_k)^p = Af_{1k} + Bf_{2k}$ with $f_{1k} = |\cos(\theta_k - \alpha)|^p$, $f_{2k} = |\sin(\theta_k - \alpha)|^p$. For fixed (p, α) the inner problem is therefore a 2-parameter linear least squares with box constraint $A, B \geq 1$, solved in closed form (interior, two edges, corner). Weighting the $y_k = c_k^p$ residuals by $1/y_k^2$ makes the weighted SSR equal $p^2 \cdot \text{SSR}_{\log}$ to first order (since $d \log c = d(c^p)/(pc^p)$), so S/p^2 is a faithful proxy for the log-scale objective and is comparable across the whole (p, α) grid. Only (p, α) is gridded (55×36 by default, $\Delta p = 0.05$, $\Delta \alpha = 2.5^\circ$), with one parabolic refinement step in each, and the sufficient statistics are five moment vectors computed as (cells $\times$ rays) matrix products — a 10^5 -cell layer fits in ~ 9 s. Both nested models are returned: the pure rotated grid ($A = B = 1$, parameters p, α) and the full model; comparing their residual σ against the axis-aligned MLE’s tells whether rotation or anisotropy is real signal in a cell. Implementation in `src/pnorm/minkowski_fit.py`; synthetic-recovery validation (noise, missing rays, degenerate cases) in `tests/test_minkowski_fit.py`.

Two new per-cell fields fall out for free: the **orientation field** $\alpha(x)$ — a continuous street-grid compass map of the city — and the **anisotropy field** $\kappa(x)$, which localizes structurally slow directions (river crossings read as low- κ bands oriented along the river). Neither is renderable from any existing output; both consume the per-ray circuitries array already saved by `circuitry_grid.py`.

Remark. At $p = 2$ the full model’s unit ball is a rotated ellipse, so the anisotropic fit contains the parent project’s Riemannian ellipse track as its $p = 2$ slice — the reconciliation anticipated in the caveats below, obtained here as a special case rather than a separate pipeline.

Caveats and future work

1. **Axis-aligned isotropy assumption.** Addressed: the preceding section generalizes $M(p)$ to a rotated, axis-weighted Minkowski functional, fitted per cell from the saved per-ray circuitries. The moment estimators turn out provably rotation-immune (the directional-circuitry distribution over a uniform ring is invariant under grid rotation); the practical bias risks are the ray-pattern MLE under rotation and *all* axis-aligned estimators under anisotropy. City-scale reruns with the new fit remain to be done.
2. **Static, all-day routing.** OSRM uses static edge weights (no congestion). Time-of-day variation matters for “real” walking and driving; the present pipeline gives a free-flow snapshot.

3. **Pedestrian network completeness.** OSRM’s foot profile relies on OSM tags. Missing sidewalks, ferry-only crossings, and pedestrian-permissive private paths all introduce noise that degrades p . NYC, with high OSM coverage, fares better than less well-mapped cities.
4. **Cross-city normalization.** Reporting p on a fixed colorbar range makes cities visually comparable, but the correct normalization for, say, “what’s the typical p at a 5-min walkshed across all U.S. metros” is an empirical question we have not yet posed.
5. **Reconciliation with the parent Riemannian track.** The sibling `src/isochrone_metric/ track` fits a Riemannian tensor $g(x)$ to an isochrone polygon and works with $p = 2$ implicitly (every unit ball is an ellipse). The two should be reconciled eventually: the Riemannian fit can be cast as the special-case $p = 2$ branch of a more general anisotropic Minkowski-norm fit that also recovers p .

Reproducibility

All inputs derive from public OSM extracts and a fixed OSRM version. The full pipeline runs locally (no rate-limited APIs). The catalog, projection switch, tile build script, OSRM Docker compose, ring-sampling code, M -inversion code, and folium renderer are all in `pnorm/`, with a running journal at `pnorm/docs/journal.md`.

For one city end-to-end:

```
CITY=<key> PROFILE=car ./scripts/build_tiles.sh # ~5 min
CITY=<key> PROFILE=foot ./scripts/build_tiles.sh # ~10 min
CITY=<key> docker compose up -d
uv run python scripts/circuitry_grid.py --city <key> \
    --spacing 250 --radius 1000 --npz data/<key>_r1000.npz
uv run python scripts/circuitry_map_multi.py --city <key> \
    --npz data/<key>_r1000.npz \
    --field effective_p --cmap spectral \
    --vmin 0.25 --vmax 1.5 \
    --out data/<key>_p.html
```